

ADVERTISING CAMPAIGN ANALYSIS

The Flamingo Grill

Case Study 4.1 · Media Budget Optimization

BACKGROUND

Setting the Stage

The Flamingo Grill is an upscale restaurant in St. Petersburg, Florida. Management engaged advertising firm Haskell & Johnson (HJ) to recommend how a \$279,000 seasonal advertising budget should be distributed across TV, radio, and online channels.

PRIMARY OBJECTIVE

Maximize total exposure ratings or new customers reached while satisfying all budget and media constraints.

\$279,000

Total Budget

100,000+

New Customers Required

3 Channels

TV · Radio · Online

Advertising Channel Overview

Channel	Exposure Rating / Ad	New Customers / Ad	Cost per Ad
Television	90	4,000	\$10,000
Radio	25	2,000	\$3,000
Online	10	1,000	\$1,000

HJ CONSULTANTS – DIMINISHING RETURNS

Television: After 10 ads, rating drops to 55 · new customers drop to 1,500
Radio: After 15 ads, rating drops to 20 · new customers drop to 1,200
Online: After 20 ads, rating drops to 5 · new customers drop to 800

Campaign Rules & Boundaries

TV

At least 2× as many radio ads as TV ads

\$

Online budget $\geq$ \$30,000

TV

No more than 20 television advertisements

Σ

Total budget $\leq$ \$279,000

\$

Television budget $\geq$ \$140,000

👥

Minimum 100,000 new customers reached

\$

Radio budget $\leq$ \$99,000

Maximize Exposure Rating

$$\text{MAX } 90t_1 + 45t_2 + 25r_1 + 20r_2 + 10o_1 + 5o_2$$

15

TV Ads

\$150,000

33

Radio Ads

\$99,000

30

Online Ads

\$30,000

2,160

Total Exposure Rating

127,100

New Customers Reached

100%

Budget Utilized

Range of Objective Coefficients

💡 A \$10,000 budget increase (\$279K → \$289K) adds 1 TV ad and raises total exposure rating by +55 (2,160 → 2,215)

Variable	Current Value	Min Range	Max Range
t ₁ (first 10 ads)	90	80	110
t ₂ (after 10 ads)	45	30	70

Radio · Moderate flexibility

r ₁ (first 15 ads)	25	20	35
r ₂ (after 15 ads)	20	10	40

Online · Least sensitive — widest ranges

o ₁ (first 20 ads)	10	5	20
o ₂ (after 20 ads)	5	0	15

Maximize New Customers Reached

$$\text{MAX } 4,000t_1 + 1,500t_2 + 2,000r_1 + 1,500r_2 + 1,000o_1 + 800o_2$$

14

TV Ads

\$140,000

28

Radio Ads

\$84,000

55

Online Ads

\$55,000

139,600

New Customers Reached

2,130

Total Exposure Rating

100%

Budget Utilized

Scenario A vs. Scenario B

Metric	Scenario A Exposure Rating	Scenario B New Customers
TV Ads	15 ads	14 ads
Radio Ads	33 ads	28 ads
Online Ads	30 ads	55 ads
TV Budget	\$150,000	\$140,000
Radio Budget	\$99,000	\$84,000
Online Budget	\$30,000	\$55,000
Exposure Rating	2,160	2,130 (-30)
New Customers	127,100	139,600 (+12,500 ✓)

FINAL RECOMMENDATION

Optimize for New Customers

The Flamingo Grill should pursue Scenario B — allocating \$140,000 to TV, \$84,000 to radio, and \$55,000 to online advertising — to maximize new customer acquisition while satisfying all constraints.

+12,500

More new customers vs. Scenario A

−30 pts

Minimal exposure rating trade-off

=\$279K

Identical total budget spent

This strategy delivers significantly better ROI at no additional cost.